

Leakage-Controlled Walk-Forward Validation of Deep Learning Models in SPY Direction Forecasting

A Case Study of Leakage Effects in Financial Machine Learning

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1 Data and Methodology

1.1 Data Description

We used daily closing price data (adjusted for splits and dividends) for the SPY ETF using the yfinance library [?]. Our sample covers 2,000 trading days from 2016-01-21 to 2023-12-29, which includes the COVID-19 crash, the recovery that followed, and the 2022 bear market. From the price series we computed daily log returns and a 20-day rolling annualized volatility. Figure ?? displays the price, returns, and volatility, and Table ?? provides the summary statistics for daily returns.

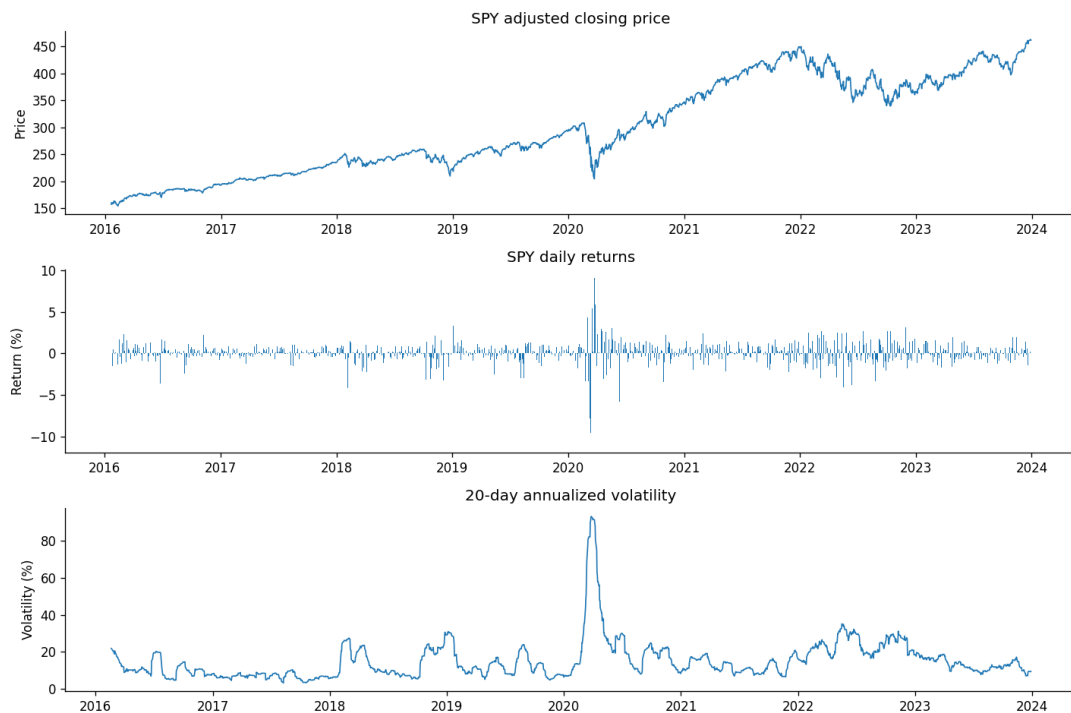


Figure 1: SPY price, daily return, and 20-day annualized volatility.

$y_t = \mathbf{1}(P_{t+20}/P_t - 1 > 0)$, i.e., whether the price 20 days later is higher than today. We chose a 20-day horizon because it balances noise and signal reasonably well. We computed forward returns before dropping incomplete windows at the end, which gave us 1,959 valid observations. As Table ?? shows, the positive class accounts for 69.10% of the labels—a clear reflection of the market’s long-term upward bias. This imbalance is important to keep in mind when interpreting accuracy and recall.

Table 2: Label distribution.

Label	Share
1 (positive 20-day return)	0.6910
0 (negative 20-day return)	0.3090

1.3 Single Train/Test Split with Three Models

We used MLP, LSTM[?], and CNN-GAF[?] in Step 1. This single split is not a clean test. It is a leaky baseline that we use for comparison.

Training observations: 1,371.

Test observations: 588.

Training period: 2016-02-22 to 2021-07-30.

Test period: 2021-08-02 to 2023-11-30.

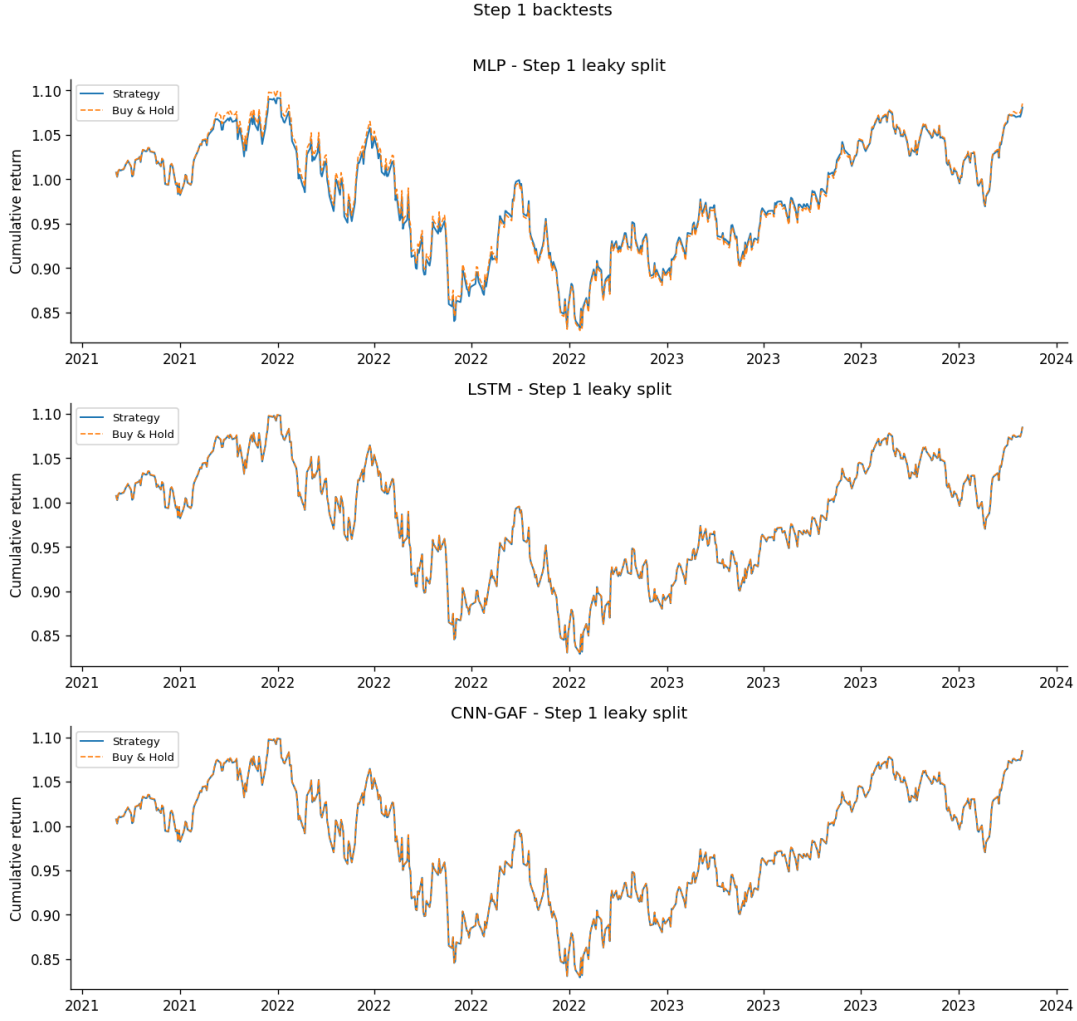


Figure 2: Step 1 cumulative returns from the leaky single split.

Table 3: Step 1 classification results.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
MLP	0.5357	0.5397	0.9812	0.6963	0.5219
LSTM	0.5425	0.5425	1.0000	0.7034	0.4863
CNN-GAF	0.5425	0.5425	1.0000	0.7034	0.4804

Table 4: Step 1 backtest results.

Model	Total Return (%)	B&H Return (%)	Sharpe	B&H Sharpe	Max DD (%)	Exposure	Trades
MLP	7.38	8.53	0.257	0.281	-23.74	0.986	15
LSTM	8.47	8.53	0.280	0.281	-24.50	1.000	1
CNN-GAF	8.47	8.53	0.280	0.281	-24.50	1.000	1

Step 1 should not be treated as a true out-of-sample test. MLP has the highest AUC-ROC, 0.5219, but its return is below buy-and-hold. LSTM and CNN-GAF stay almost fully long, so their returns are close to buy-and-hold.

This step is still useful. It shows how a weak validation design can make a result look more useful than it is.

2 Walk-Forward Backtesting Without Embargo

2.1 Step 2a: 500/500 Walk-Forward

Figure ?? shows the cumulative returns from the no-embargo 500/500 walk-forward, and Tables ?? and ?? give the classification and backtest metrics.

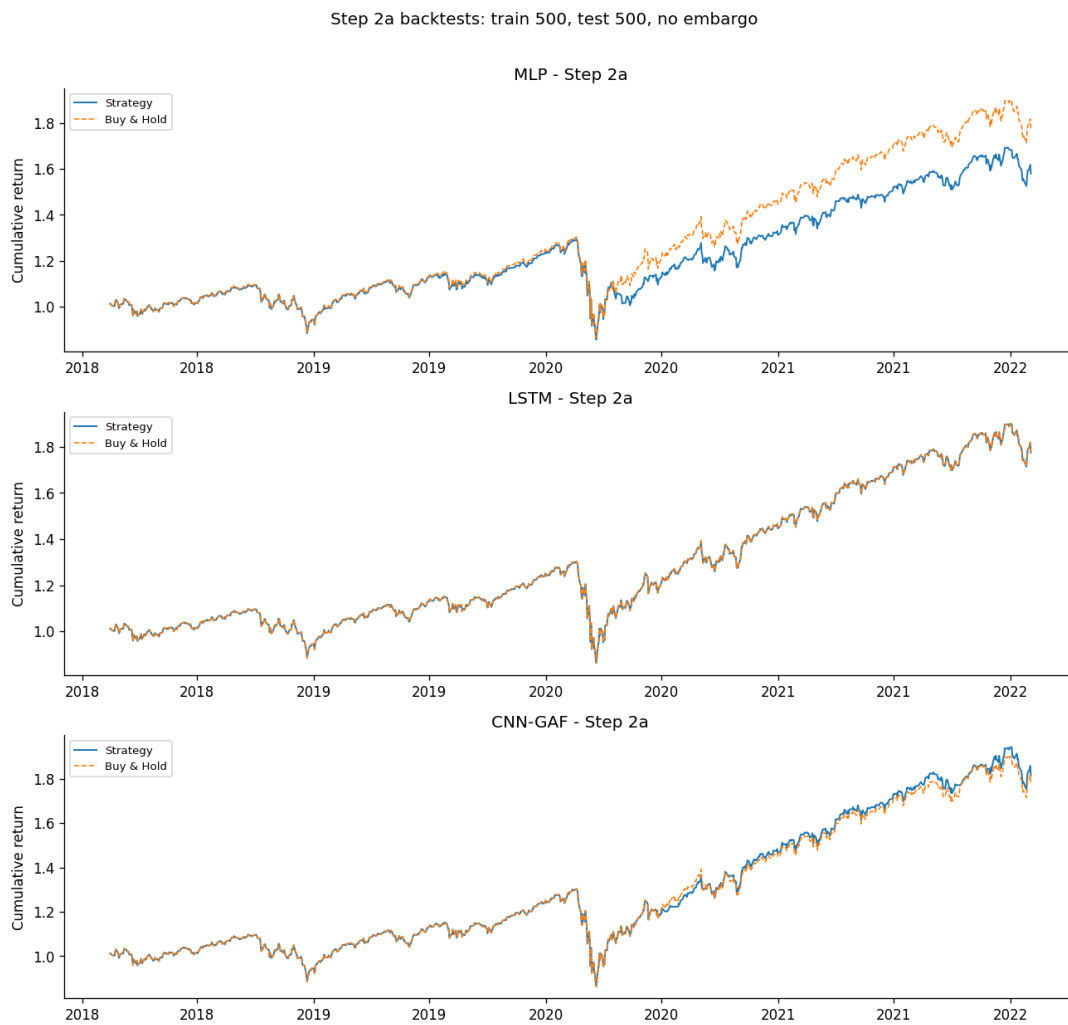


Figure 3: Step 2a cumulative returns (500/500, no embargo).

Table 5: Step 2a classification results.

Model	Accuracy	Precision	Recall	F1	AUC-ROC	Folds	Test Obs.
MLP	0.6870	0.7098	0.9510	0.8129	0.4623	2	1000
LSTM	0.7150	0.7150	1.0000	0.8338	0.4530	2	1000
CNN-GAF	0.6840	0.7076	0.9510	0.8115	0.4822	2	1000

Table 6: Step 2a backtest results.

Model	Total Return (%)	B&H Return (%)	Sharpe	B&H Sharpe	Max DD (%)	Exposure	Trades
MLP	57.99	77.55	0.668	0.801	-33.72	0.958	75
LSTM	77.46	77.55	0.800	0.801	-33.72	1.000	1
CNN-GAF	81.41	77.55	0.835	0.801	-33.72	0.961	27

Even without an embargo, all three models produced AUC-ROC values below 0.50—essentially worse than random guessing—once we removed the scaling leak. Yet CNN-GAF still delivered an 81.41% return, beating buy-and-hold’s 77.55%. At first glance that looks attractive, but it is largely explained by the model being invested 96.1% of the time during a strong bull market. The LSTM went one step further: it entered a long position at the start and never closed it (exposure 1.0, only one trade), so its returns almost perfectly mirrored the benchmark. The MLP, despite its high accuracy and precision, only managed 57.99%, lagging the market by nearly 20 percentage points. With only two folds, these numbers are extremely sensitive to the particular time periods chosen; we should not overinterpret CNN-GAF’s small outperformance here.

2.2 Step 2b: 500/100 Walk-Forward

Using a shorter test window and retraining more frequently (14 folds) improved the classification metrics for LSTM and CNN-GAF, as shown in Figure ?? and Tables ??–??.



Figure 4: Step 2b cumulative returns (500/100, no embargo).

Table 7: Step 2b classification results.

Model	Accuracy	Precision	Recall	F1	AUC-ROC	Folds	Test Obs.
MLP	0.5836	0.6602	0.7701	0.7110	0.5048	14	1400
LSTM	0.6371	0.6618	0.9291	0.7730	0.5239	14	1400
CNN-GAF	0.6586	0.6628	0.9903	0.7941	0.5842	14	1400

Table 8: Step 2b backtest results.

Model	Total Return (%)	B&H Return (%)	Sharpe	B&H Sharpe	Max DD (%)	Exposure	Trades
MLP	4.91	81.35	0.141	0.621	-36.14	0.776	275
LSTM	66.38	81.35	0.562	0.621	-33.72	0.934	65
CNN-GAF	82.66	81.35	0.629	0.621	-33.72	0.994	5

CNN-GAF produced the strongest numbers here: AUC-ROC 0.5842 and a return of 82.66%, just above buy-and-hold (81.35%). LSTM was more modest at 0.5239, while MLP stayed near random (0.5048) and delivered

a paltry 4.91% return. It seems that more frequent retraining might have helped CNN-GAF pick up on shorter-term patterns, but we cannot trust this edge yet. Because there is no embargo, the 20-day labels at the fold boundaries still leak into the test set, so these results are likely inflated.

2.3 Step 2 Compared with Step 1

Walk-forward testing is more realistic than the single split because it keeps time order. It also fits the scaler inside each training fold, so test data are not used for scaling. The results become less clean. A high return does not always mean better prediction. In Step 2a, AUC-ROC is weak, but returns are high because SPY rises.

2.4 Step 2a Compared with Step 2b

Step 2b gives better AUC-ROC for LSTM and CNN-GAF. CNN-GAF moves from 0.4822 in Step 2a to 0.5842 in Step 2b. More frequent retraining may help the model adjust to the market.

However, Step 2 still has boundary leakage from the 20-day forward labels. This can make the results look better than they are. Step 3 adds an embargo to reduce this problem.

3 Walk-Forward Backtesting With Embargo

3.1 Embargo Implementation

To eliminate the boundary leakage, we introduced a 20-day embargo gap [?] between the training and test sets. For Step 3b we used 500 days for training, 20 days embargo, and 500 days for testing (2 folds). For Step 3c we used 500/20/100 (14 folds).

3.2 Step 3b: 500/500 Walk-Forward with Embargo

Figure ?? and Tables ??–?? show what happened after we applied the embargo.

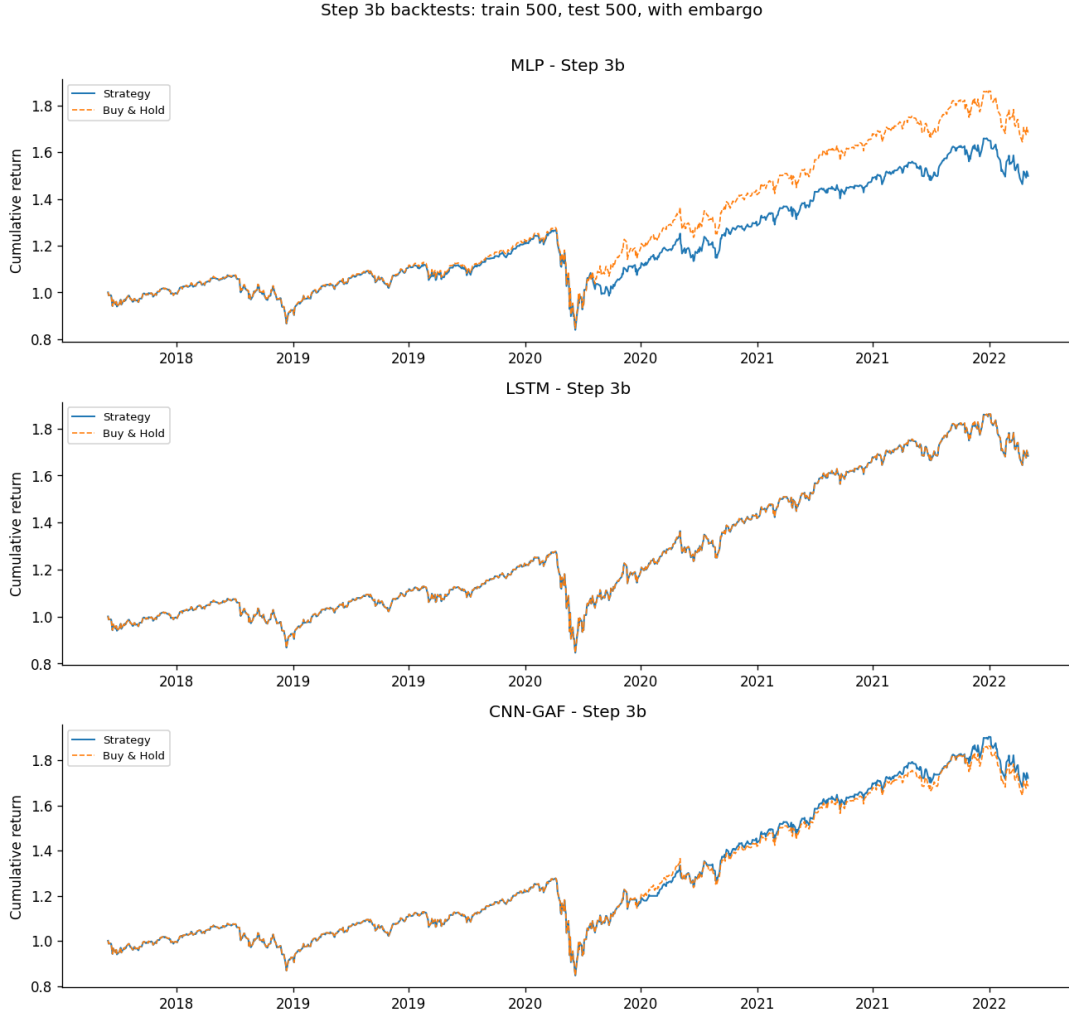


Figure 5: Step 3b cumulative returns (500/500, with embargo).

Table 9: Step 3b classification results.

Model	Accuracy	Precision	Recall	F1	AUC-ROC	Folds	Test Obs.
MLP	0.6930	0.7161	0.9515	0.8172	0.4666	2	1000
LSTM	0.7210	0.7210	1.0000	0.8379	0.4363	2	1000
CNN-GAF	0.6900	0.7138	0.9515	0.8157	0.4508	2	1000

Table 10: Step 3b backtest results.

Model	Total Return (%)	B&H Return (%)	Sharpe	B&H Sharpe	Max DD (%)	Exposure	Trades
MLP	49.87	68.43	0.601	0.734	-33.72	0.958	75
LSTM	68.35	68.43	0.733	0.734	-33.72	1.000	1
CNN-GAF	72.09	68.43	0.767	0.734	-33.72	0.961	27

After we introduced the embargo, our perspective changed. Every AUC-ROC dropped below 0.50—clear evidence that none of the models could distinguish between up and down months. The LSTM’s perfect recall (1.0) turned out to

be a mechanical artifact: it almost always predicted “up,” simply exploiting the class imbalance rather than learning anything meaningful. CNN-GAF’s 72.09% return beat buy-and-hold by 3.66 percentage points, but we noticed that its market exposure was 96.1%, so the excess came from trend participation, not from skilful timing. The combination of positive returns and near-random AUC-ROC shows that a blind long bias can look profitable in a rising market, and with only two folds, these embargo results are fragile—a small shift in the boundary could easily flip the outcome.

3.3 Step 3c: 500/100 Walk-Forward with Embargo

Our cleanest protocol, Step 3c, increases the number of folds to 14, giving us a more detailed view. Results are shown in Figure ?? and Tables ??–??.



Figure 6: Step 3c cumulative returns (500/100, with embargo).

Table 11: Step 3c classification results.

Model	Accuracy	Precision	Recall	F1	AUC-ROC	Folds	Test Obs.
MLP	0.5821	0.6581	0.7694	0.7094	0.5005	14	1400
LSTM	0.6357	0.6579	0.9386	0.7735	0.5301	14	1400
CNN-GAF	0.6486	0.6580	0.9784	0.7868	0.5709	14	1400

Table 12: Step 3c backtest results.

Model	Total Return (%)	B&H Return (%)	Sharpe	B&H Sharpe	Max DD (%)	Exposure	Trades
MLP	-7.36	72.01	0.023	0.575	-36.19	0.775	269
LSTM	52.84	72.01	0.482	0.575	-33.72	0.946	59
CNN-GAF	71.03	72.01	0.572	0.575	-33.72	0.986	9

Under our most rigorous test, the picture became clearer. CNN-GAF achieved the best AUC-ROC (0.5709), but that is still well short of the 0.70 level that is typically considered useful. Its return (71.03%) barely lagged buy-and-hold (72.01%), and the Sharpe ratios were almost identical. LSTM only managed 0.5301 AUC-ROC and a return of 52.84%, well behind the benchmark. MLP was essentially random (AUC-ROC 0.5005) and even lost money (-7.36%), though its accuracy and precision did not look dramatically worse. The models that stayed heavily long—CNN-GAF and LSTM—simply rode the market’s upward drift; their benchmark-like returns evaporated once we stripped out the trend. This confirms that none of the architectures could reliably predict SPY’s 20-day direction.

3.4 Comparison After the Embargo: 3b vs. 3c and Overfitting Assessment

3.4.1 Step 3b versus Step 3c

Increasing the test frequency from 500 to 100 observations raises the number of folds from 2 to 14, improving statistical robustness. While AUC-ROC improves marginally—CNN-GAF rises from 0.4508 in Step 3b to 0.5709 in Step 3c—all values remain well below the 0.70 threshold for useful prediction. More importantly, the excess return over buy-and-hold evaporates; CNN-GAF goes from outperforming by 3.66% in Step 3b to underperforming by 0.98% in Step 3c. The shorter test windows in Step 3c reduce the artificial edge of a persistent long bias during the bullish period.

3.4.2 Has Overfitting Disappeared? (Step 3 vs. Step 2)

Comparing the embargoed results to Step 2 reveals that overfitting has not disappeared—it has merely been exposed. For example, CNN-GAF’s Step

2b excess return over B&H (1.31%, with 0.5842 AUC-ROC) turns into a -0.98% shortfall in Step 3c, despite a similar AUC-ROC (0.5709). The LSTM’s perfect recall in Step 2 is further revealed as a mechanical artifact of majority-class prediction. The embargo strips away the boundary leakage that inflated past performance, showing that the models lack genuine directional foresight. The positive returns that remain are driven by market drift, not predictive skill. Consequently, overfitting is still present, but the embargo provides a more honest diagnostic framework.

4 Output Summary

Tables ?? and ?? bring together all classification and backtest results across the four protocols.

Table 13: Consolidated classification results across all validation protocols.

Protocol	Model	Accuracy	Precision	Recall	F1	AUC-ROC	Folds	Test Obs.
Step 1 – leaky single split	MLP	0.5357	0.5397	0.9812	0.6963	0.5219	–	–
Step 1 – leaky single split	LSTM	0.5425	0.5425	1.0000	0.7034	0.4863	–	–
Step 1 – leaky single split	CNN-GAF	0.5425	0.5425	1.0000	0.7034	0.4804	–	–
Step 2a – WF 500/500 no embargo	MLP	0.6870	0.7098	0.9510	0.8129	0.4623	2	1000
Step 2a – WF 500/500 no embargo	LSTM	0.7150	0.7150	1.0000	0.8338	0.4530	2	1000
Step 2a – WF 500/500 no embargo	CNN-GAF	0.6840	0.7076	0.9510	0.8115	0.4822	2	1000
Step 2b – WF 500/100 no embargo	MLP	0.5836	0.6602	0.7701	0.7110	0.5048	14	1400
Step 2b – WF 500/100 no embargo	LSTM	0.6371	0.6618	0.9291	0.7730	0.5239	14	1400
Step 2b – WF 500/100 no embargo	CNN-GAF	0.6586	0.6628	0.9903	0.7941	0.5842	14	1400
Step 3b – WF 500/500 with embargo	MLP	0.6930	0.7161	0.9515	0.8172	0.4666	2	1000
Step 3b – WF 500/500 with embargo	LSTM	0.7210	0.7210	1.0000	0.8379	0.4363	2	1000
Step 3b – WF 500/500 with embargo	CNN-GAF	0.6900	0.7138	0.9515	0.8157	0.4508	2	1000
Step 3c – WF 500/100 with embargo	MLP	0.5821	0.6581	0.7694	0.7094	0.5005	14	1400
Step 3c – WF 500/100 with embargo	LSTM	0.6357	0.6579	0.9386	0.7735	0.5301	14	1400
Step 3c – WF 500/100 with embargo	CNN-GAF	0.6486	0.6580	0.9784	0.7868	0.5709	14	1400

Table 14: Consolidated backtest results across all validation protocols.

Protocol	Model	Total Return (%)	B&H Return (%)	Sharpe	B&H Sharpe	Max DD (%)	Exposure	Trades
Step 1 – leaky single split	MLP	7.38	8.53	0.257	0.281	-23.74	0.986	15
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Step 2a – WF 500/500 no embargo	CNN-GAF	81.41	77.55	0.835	0.801	-33.72	0.961	27
Step 2b – WF 500/100 no embargo	MLP	4.91	81.35	0.141	0.621	-36.14	0.776	275
Step 2b – WF 500/100 no embargo	LSTM	66.38	81.35	0.562	0.621	-33.72	0.934	65
Step 2b – WF 500/100 no embargo	CNN-GAF	82.66	81.35	0.629	0.621	-33.72	0.994	5
Step 3b – WF 500/500 with embargo	MLP	49.87	68.43	0.601	0.734	-33.72	0.958	75
Step 3b – WF 500/500 with embargo	LSTM	68.35	68.43	0.733	0.734	-33.72	1.000	1
Step 3b – WF 500/500 with embargo	CNN-GAF	72.09	68.43	0.767	0.734	-33.72	0.961	27
Step 3c – WF 500/100 with embargo	MLP	-7.36	72.01	0.023	0.575	-36.19	0.775	269
Step 3c – WF 500/100 with embargo	LSTM	52.84	72.01	0.482	0.575	-33.72	0.946	59
Step 3c – WF 500/100 with embargo	CNN-GAF	71.03	72.01	0.572	0.575	-33.72	0.986	9

5 Conclusion

As we tightened the validation process—from a leaky split to a purged-embargo walk-forward—the apparent edge gradually disappeared. Under our clean-

est protocol, CNN-GAF’s AUC-ROC reached only 0.5709, and its return (71.03%) barely trailed buy-and-hold (72.01%). None of the models showed statistical or economic significance.

We draw a few clear lessons. First, leaky protocols inflate metrics; without an embargo, high market exposure can easily mask the absence of directional skill. Second, once the embargo removed all leakage, every model simply rode the market drift—AUC-ROC near or below 0.5 and benchmark –like returns confirmed that there was no genuine predictive signal. Third, the validation design turned out to be just as critical as the architecture choice; the purged-embargo method [?] should be adopted as the standard practice.

For future work, any improvements in features, labeling or regime-aware modeling must be tested under the same rigorous, leakage-free framework. For now, none of the architectures we examined warrants deployment in a live trading environment.

References

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